

Project1

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2024-11-22

This data is about the sales of video games. The goal of this project is to identify what are the most popular genres of games that were sold in each region, and possibly predict the sales for future games.

```
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
library(ggplot2)
library(tidyr)
video_games <- read.csv("Video_Games.csv")
summary(video_games)
```

```
##      Name      Platform      Year_of_Release      Genre
## Length:16719 Length:16719 Length:16719 Length:16719
## Class :character Class :character Class :character Class :character
## Mode :character Mode :character Mode :character Mode :character
##
##
##
##
## Publisher      NA_Sales      EU_Sales      JP_Sales
## Length:16719 Min. : 0.0000 Min. : 0.000 Min. : 0.0000
## Class :character 1st Qu.: 0.0000 1st Qu.: 0.000 1st Qu.: 0.0000
## Mode :character Median : 0.0800 Median : 0.020 Median : 0.0000
## Mean : 0.2633 Mean : 0.145 Mean : 0.0776
## 3rd Qu.: 0.2400 3rd Qu.: 0.110 3rd Qu.: 0.0400
## Max. :41.3600 Max. :28.960 Max. :10.2200
##
## Other_Sales      Global_Sales      Critic_Score      Critic_Count
## Min. : 0.00000 Min. : 0.0100 Min. :13.00 Min. : 3.00
## 1st Qu.: 0.00000 1st Qu.: 0.0600 1st Qu.:60.00 1st Qu.: 12.00
## Median : 0.01000 Median : 0.1700 Median :71.00 Median : 21.00
## Mean : 0.04733 Mean : 0.5335 Mean :68.97 Mean : 26.36
## 3rd Qu.: 0.03000 3rd Qu.: 0.4700 3rd Qu.:79.00 3rd Qu.: 36.00
## Max. :10.57000 Max. :82.5300 Max. :98.00 Max. :113.00
##
## NA's :8582 NA's :8582
```

```
## User_Score      User_Count      Developer      Rating
## Length:16719    Min.      :    4.0    Length:16719    Length:16719
## Class :character 1st Qu.:   10.0    Class :character Class :character
## Mode  :character Median :   24.0    Mode  :character Mode  :character
##                Mean   :  162.2
##                3rd Qu.:   81.0
##                Max.   :10665.0
##                NA's   :9129
```

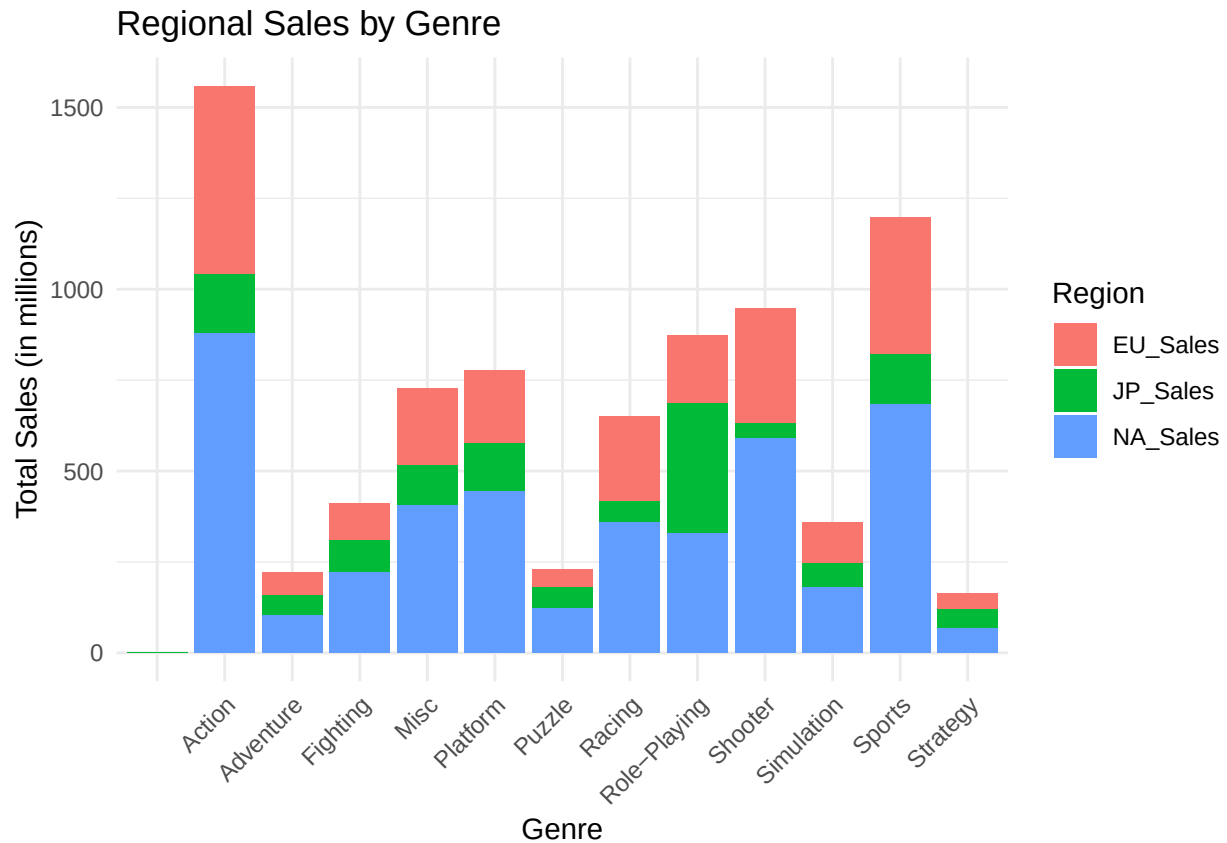
```
head(video_games)
```

```
##              Name Platform Year_of_Release      Genre Publisher
## 1           Wii Sports      Wii          2006      Sports  Nintendo
## 2      Super Mario Bros.      NES          1985 Platform  Nintendo
## 3      Mario Kart Wii      Wii          2008      Racing  Nintendo
## 4      Wii Sports Resort      Wii          2009      Sports  Nintendo
## 5 Pokemon Red/Pokemon Blue      GB          1996 Role-Playing Nintendo
## 6              Tetris      GB          1989      Puzzle  Nintendo
##  NA_Sales EU_Sales JP_Sales Other_Sales Global_Sales Critic_Score Critic_Count
## 1    41.36   28.96    3.77         8.45      82.53           76           51
## 2    29.08    3.58    6.81         0.77      40.24           NA           NA
## 3    15.68   12.76    3.79         3.29      35.52           82           73
## 4    15.61   10.93    3.28         2.95      32.77           80           73
## 5    11.27    8.89   10.22         1.00      31.37           NA           NA
## 6    23.20    2.26    4.22         0.58      30.26           NA           NA
##  User_Score User_Count Developer Rating
## 1          8        322  Nintendo      E
## 2          NA
## 3         8.3        709  Nintendo      E
## 4          8        192  Nintendo      E
## 5          NA
## 6          NA
```

I want to only see the sales of NA, EU, and JP for this project, as those summarize a good sample of sales.

```
# Choose only specified columns
video_games <- video_games %>%
  select(Year_of_Release, Genre, NA_Sales, EU_Sales, JP_Sales)
str(video_games)
```

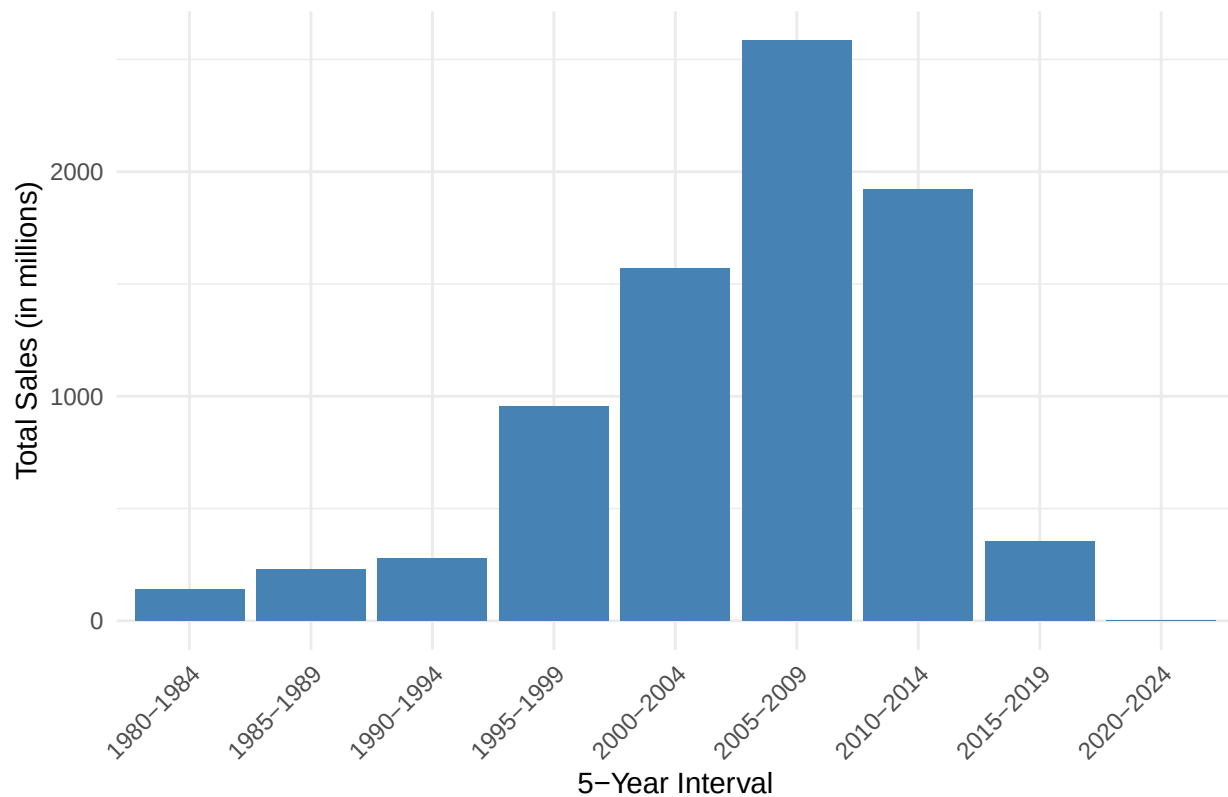
```
## 'data.frame': 16719 obs. of 5 variables:
## $ Year_of_Release: chr "2006" "1985" "2008" "2009" ...
## $ Genre : chr "Sports" "Platform" "Racing" "Sports" ...
## $ NA_Sales : num 41.4 29.1 15.7 15.6 11.3 ...
## $ EU_Sales : num 28.96 3.58 12.76 10.93 8.89 ...
## $ JP_Sales : num 3.77 6.81 3.79 3.28 10.22 ...
```



As we can see here that NA region had the most sales but considering the population that is not a surprise. Now in the NA region, Action genre was most popular as well as in the EU region. In the JP region, Role-Playing was the most popular.

```
## Warning: There was 1 warning in `mutate()`.
## i In argument: `Year_of_Release = as.numeric(as.character(Year_of_Release))`.
## Caused by warning:
## ! NAs introduced by coercion
```

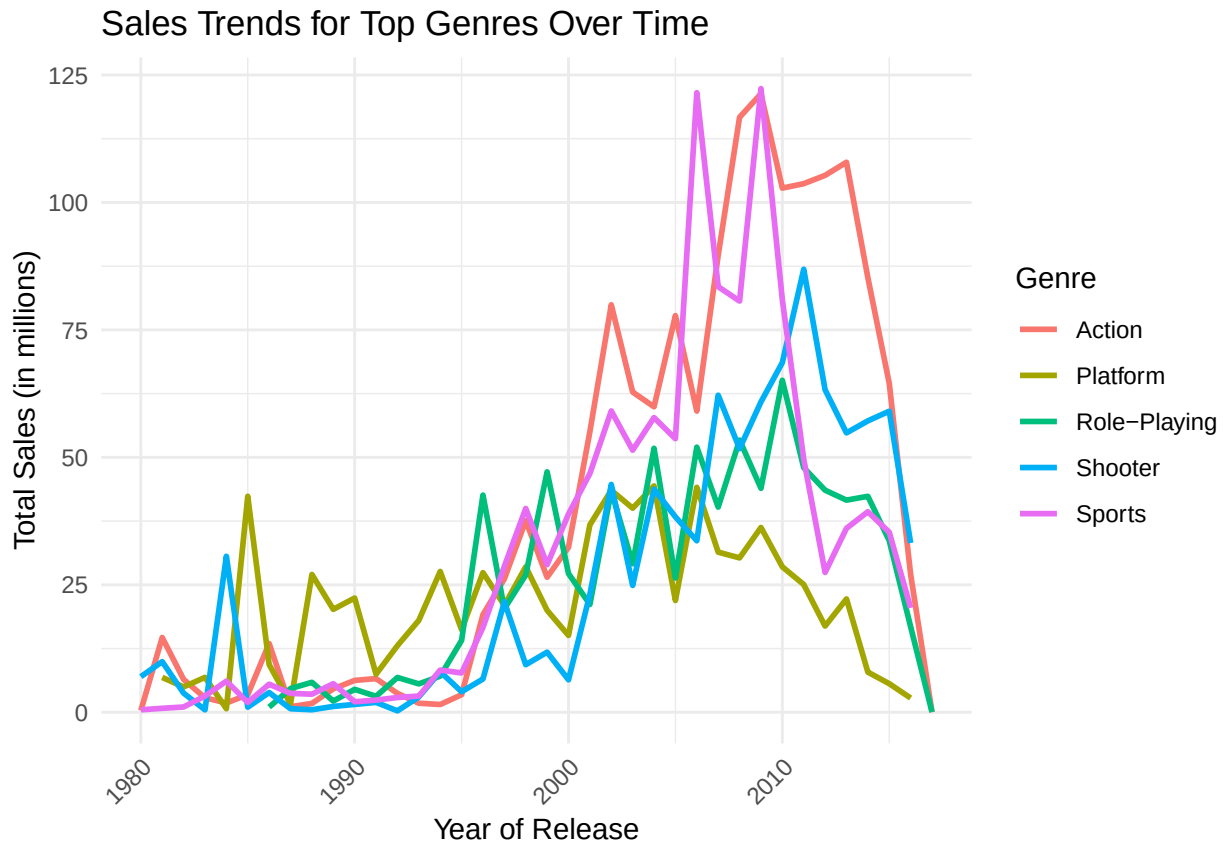
Video Game Sales by 5-Year Intervals



Here I wanted to see if the release of New consoles have a direct impact on the number of sales. Here it does show some correlation as most of the consoles that run the video games were released post 1995, and after the PS4 released in 2013, we can see the trend of video games decreasing as the PS5 and Xbox Series X weren't released until 2020.

```
## `summarise()` has grouped output by 'Year_of_Release'. You can override using  
## the `groups` argument.
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.  
## i Please use `linewidth` instead.  
## This warning is displayed once every 8 hours.  
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was  
## generated.
```



Here is a graph of the trends of video game sales over time, we can see that the genres action and sports have the biggest spikes with shooter genre having a fair bit of spike.

Now I will create a linear regression model using the trends of Na, EU, and JP regions to predict the sales trend

```
# Aggregate regional sales by year
sales_by_year <- video_games %>%
  group_by(Year_of_Release) %>%
  summarise(
    NA_Sales = sum(NA_Sales, na.rm = TRUE),
    EU_Sales = sum(EU_Sales, na.rm = TRUE),
    JP_Sales = sum(JP_Sales, na.rm = TRUE)
  ) %>%
  filter(Year_of_Release >= 1980) # Focus on years from 1980 onward

# Building separate linear regression models for each region
na_model <- lm(NA_Sales ~ Year_of_Release, data = sales_by_year)
eu_model <- lm(EU_Sales ~ Year_of_Release, data = sales_by_year)
jp_model <- lm(JP_Sales ~ Year_of_Release, data = sales_by_year)

# Forecast sales for the next 10 years for each region
future_years <- data.frame(Year_of_Release = seq(max(sales_by_year$Year_of_Release) + 1,
                                                  max(sales_by_year$Year_of_Release) + 10))

future_years <- future_years %>%
  mutate(
    NA_Sales = predict(na_model, newdata = future_years),
    EU_Sales = predict(eu_model, newdata = future_years),
    JP_Sales = predict(jp_model, newdata = future_years)
  )
```

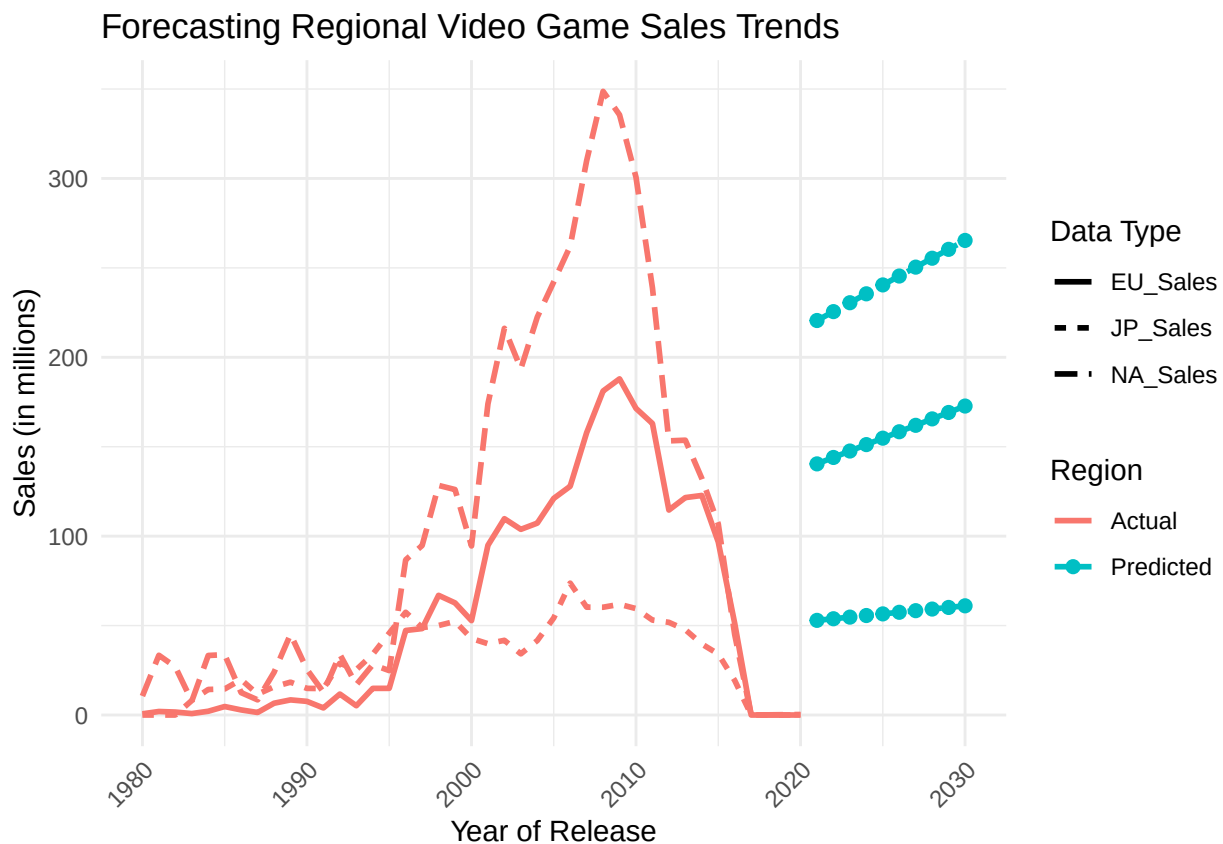
```

)

# Combine actual and predicted data for visualization
combined_data <- sales_by_year %>%
  mutate(Type = "Actual") %>%
  bind_rows(
    future_years %>%
      mutate(Type = "Predicted")
  ) %>%
  pivot_longer(cols = c(NA_Sales, EU_Sales, JP_Sales), names_to = "Region", values_to = "Sales")

# Visualizing the results
ggplot(combined_data, aes(x = Year_of_Release, y = Sales, color = Type, linetype = Region)) +
  geom_line(size = 1) +
  geom_point(data = filter(combined_data, Type == "Predicted"), size = 2) +
  theme_minimal() +
  labs(
    title = "Forecasting Regional Video Game Sales Trends",
    x = "Year of Release",
    y = "Sales (in millions)",
    color = "Region",
    linetype = "Data Type"
  ) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

```



```

# Prepare data
model_data <- video_games %>%

```

```

filter(!is.na(Genre), Year_of_Release >= 1980) %>%
select(Genre, NA_Sales, EU_Sales, JP_Sales, Year_of_Release) %>%
mutate(Genre = as.factor(Genre))

# Aggregate data by Genre and Region
genre_region_sales <- model_data %>%
  group_by(Year_of_Release, Genre) %>%
  summarise(
    NA_Sales = sum(NA_Sales, na.rm = TRUE),
    EU_Sales = sum(EU_Sales, na.rm = TRUE),
    JP_Sales = sum(JP_Sales, na.rm = TRUE)
  ) %>%
  ungroup()

## `summarise()` has grouped output by 'Year_of_Release'. You can override using
## the ``.groups` argument.

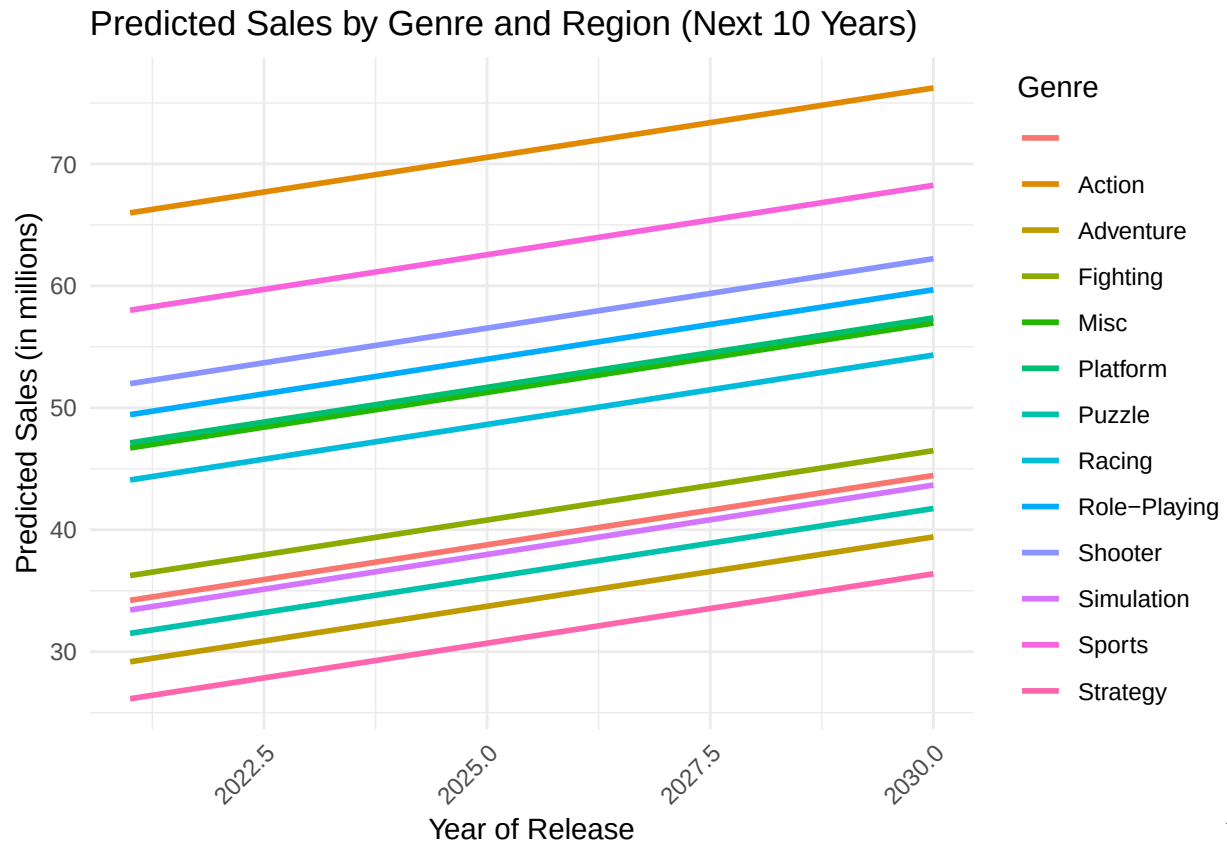
# linear regression model
sales_model <- lm(NA_Sales + EU_Sales + JP_Sales ~ Year_of_Release + Genre, data = genre_region_sales)

# Predicting sales for the next 10 years by genre and region
future_years <- expand_grid(
  Year_of_Release = seq(max(genre_region_sales$Year_of_Release) + 1,
                        max(genre_region_sales$Year_of_Release) + 10),
  Genre = unique(genre_region_sales$Genre)
)

future_sales <- future_years %>%
  mutate(
    Predicted_Sales = predict(sales_model, newdata = future_years)
  )

# Visualizing the results
ggplot(future_sales, aes(x = Year_of_Release, y = Predicted_Sales, color = Genre)) +
  geom_line(size = 1) +
  theme_minimal() +
  labs(
    title = "Predicted Sales by Genre and Region (Next 10 Years)",
    x = "Year of Release",
    y = "Predicted Sales (in millions)",
    color = "Genre"
  ) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

```



We can conclude that factors like different genres of video games and the popularity of those genres in different regions might persuade game developers to develop more games in trending genres, and possibly make more sales in specific regions targetting people in those regions.